a) Annotated Checklist

GodBox Demo Plan

a) Annotated checklist

1. Preventative guidelines before the demo starts

Inform Sequentiality

How this applies to GodBox:
Because GodBox combines a physical sandbox and XR visuals, the participant needs to understand the order of events before putting on the headset. This reduces confusion and stops them from rushing ahead.

☑ **Explain the full sequence from start to finish**
The participant is told that the demo will begin with a short briefing of the phases in game, then headset donning, then exploration and shaping of the sandbox world. The participant will then be asked if they are ready to go to the next phase while with the headset donned. Then, a short ending sequence where they remove the headset and discuss the experience.

☑ **Clarify when instructions will be given**
Most instructions are given before headset use, with only short reminders during the experience to avoid overload and overwhelm.

☑ **Explain how to start and how to exit the experience**
The participant is told where to stand, how to begin interacting, and that they can stop at any time by verbally saying they want to stop or by pausing and raising a hand.

☑ **Describe what all facilitators will do during the demo**
One facilitator introduces the experience and gives instruction, monitoring safety, headset fit, headset link (via cable) and position around the sandbox. The other facilitator will be in charge of the computer and sequencing of the game, watching the participants gameplay and the projection on the sand.

☑ **Repeat key steps after the HMD is donned**
Once the headset is on, the facilitator briefly repeats the first action: look at the sandbox, orient to the scene, and begin shaping the sand slowly. Ask how it feels, and make sure the participant is feeling comfortable in this XR world.

Inform About Differences in Interactions & Technologies

How this applies to GodBox:
Participants may assume GodBox works like a normal VR game with controllers, menus, or hand tracking. That mismatch could easily cause breakdowns.

☑ **Ask briefly about prior VR experience**
Participants are asked whether they have used VR before.

☑ **Explicitly state what makes this setup different**
They are told that GodBox is not a typical controller-based VR game, but an embodied experience tied to a physical sandbox; That it connects the physical to the digital.

☑ **Describe interaction methods step by step**
The facilitator explains that interaction happens mainly by shaping the sand and observing how the virtual system responds in the first phase, and in the second phase, they will be shaping "society" by using a menu to pick a choose objects they want in their world. They can use hand gestures to poke buttons on the menu via direct interaction, or use raycasting for indirect. The menu and platform that spawns objects is moveable, so they can decide for themselves if they want it close to their body and use direct interaction, or further away for heavier indirect interaction usage.

☑ **Explain how errors or mistakes can be corrected**
Participants are told that there is no "wrong" action, and that if they become confused the facilitator will help reorient them. If they have picked an object they no longer want, they can "throw it into the void" (The space below them.)

☑ **Normalize confusion**
The facilitator explains that the setup is unusual and that it may take a few seconds to understand how physical shaping and virtual feedback connect. The participant can feel free to take their time to adjust to the setting and feel comfortable, there is no time limitations to this game.

Preempt the Physical Surroundings

☑ **Describe the physical area before HMD donning**
The participant is shown the sandbox, surrounding floor area, How the headset is connected (via link, which will mean they must stay somewhat static), where the RealSense camera is and how it works (and raise spatial awareness so to try to avoid collisions with it), and where they should stand for the best experience.

☑ **Explain boundaries and safe movement zones**
The facilitator identifies the safe standing area and explains that the participant should remain in front of the sandbox rather than stepping around it.

☑ **Clarify how virtual movement maps to physical movement**
Participants are told that their main movement is small bodily repositioning around the sandbox, not walking freely through the room.

☑ **Warn about bystanders and facilitators moving nearby**
The participant is informed that facilitators may move slightly nearby for safety or technical support, but that we will mainly use our voices to communicate with them.

☑ **Use tactile or visible floor markers when possible**
Tape markers/Post its on the floor indicate where the participant should stand relative to the sandbox.

Negotiate Touch

☑ **Explain that touch may be used for safety or guidance**
Participants are informed that a facilitator may lightly touch their shoulder or arm if they drift too far or need help orienting.

☑ **Specify who may initiate touch**
Only facilitators may use touch, never spectators. The experience is to be played solo.

☑ **Ask for explicit consent before starting**
Consent is requested before the headset is donned.

☑ **Offer alternatives**
The participant is told they can request voice-only guidance if needed, but we may ask them to remove the headset if they are physically drifting.

☑ **Reconfirm consent if the situation escalates**
If stronger physical guidance is needed, the facilitator asks again before doing so.

2. Alignment and repair during the demonstration

Procedural Alignment

☑ **Verbally restate the current step**
The facilitator says things like, “Right now you’re exploring how the world responds to the sand.” Ask how it feels, and if they are ready for the next step.

☑ **Explicitly mark transitions**
The facilitator signals moments such as “Now try shaping the hill,” (Phase 1) or “Next we’ll observe what happens when you place a house.” (Phase 2)

☑ **Pause the experience if possible to re-explain the sequence**
If confusion arises, the facilitator pauses and gives a brief restatement of the step and what to do.

☑ **Clarify how the participant can signal they want to stop**
The participant is reminded they can say “stop” at any time, and to let us know if anything seems off/wrong.

☑ **Re-align on who is responsible for the next action**
The facilitator clarifies whether the participant should act or wait.

Technological Alignment

☑ **Explicitly state which interactions are supported**
Participants are told that shaping the sand changes the environment, and that height/depth can cause topography changes, but not every possible gesture or object will do something.

☑ **Name what will not work**
The facilitator explains that this experience is a prototype, and some hand gestures/object selections may not work fault free.

☑ **Describe what feedback to expect**
Participants are told to expect visual changes in the world and, if relevant, projection, sound, or environmental feedback.

☑ **Acknowledge prior experience and distinguish this setup from others**
The facilitator says that previous VR experience may help with comfort, but not necessarily with understanding this particular interaction logic. The facilitator can explicitly explain interactions again upon participant confusion,

☑ **Demonstrate the interaction before donning HMD**
A short example is shown using the sandbox itself or an external display.
The projection mapping of the topography onto the sandbox gives spectators a vague idea of what they can expect in the digital experience.

Conversational Alignment

☑ **Ask open check-ins**
Questions such as “What do you see changing?” or “How is it responding?” help reveal understanding.

☑ **Repeat or rephrase questions if no response**
If the participant is silent, the facilitator simplifies the question after a pause, or asks if the participant is feeling ok in the experience.

☑ **Move closer to ensure audibility**
The facilitator positions themselves so instructions can be heard clearly.

☑ **Use confirmation questions**
For example: “Are you standing in front of the sandbox now?” or “Can you see the landscape update?”

☑ **Let the participant narrate their actions**
Participants may be asked to briefly describe what they are doing and seeing.

Visual Alignment

☑ **Experience the VR content before handing over the HMD**
The facilitator tests the scene and interaction beforehand.

☑ **Ask the participant to describe what they see**
This helps identify whether calibration or orientation is off.

☑ **Allow the participant to briefly remove the HMD if needed**
If disorientation occurs, they can recheck the physical sandbox.

☑ **Use external screens to cast the HMD view if available**
A cast view lets facilitators monitor what the participant is seeing.

Physical Alignment

☑ **Ask for permission before touching**
The facilitator requests permission before guiding physically.

☑ **Use the lightest touch necessary**
Touch is limited to hand, shoulder or elbow guidance. If necessary, help with the headset.

☑ **Combine touch with verbal explanation**
For example, “I’m just moving you half a step left so you’re centred with the box.”

☑ **Be attentive to participant comfort and hesitation**
Any discomfort is treated as a cue to stop.

☑ **Stop immediately if discomfort is expressed or sensed**
Voice-only guidance is used instead.

3. Bobbe et al. design principles applied to GodBox

Communication goals

Context
☑ GodBox is embedded in a recognisable activity: shaping a landscape with one's hands.
☑ The sandbox/topography with projection mapping metaphor makes the purpose easy to understand.
☑ The context is playful and accessible for non-experts.
☑ However, the presentation should avoid making the system look more finished or technically robust than it really is.

Show the Invisible
☑ Visual changes in the virtual world reveal otherwise invisible sensing and computation.
☑ Projection or external display can help observers see what the participant is affecting.

☑ Auditory cues will help participants understand the selection and placement process during phase 2 of the experience.

Authenticity
☑ Real hardware, sensing, and system behaviour are part of the demonstration.
☑ Facilitators should be transparent if any element is simulated, delayed, or partially staged.
(If some features are not fully stable, the facilitator should state this openly, although switching "phases" may be partially Wizard-of-Oz style. That maintains trust and supports reflection on the system as a research prototype.)

Attraction goals

Relatedness
☑ The sandbox is familiar from childhood play, model-making, and terrain sculpting.
☑ The idea of "building a world" is easy to connect to games and imagination.

☑ The projection mapping may raise curiosity about the experience, adn the tactile quality of the sand.

Eye-Catcher
☑ The physical sandbox, XR visuals, and changing landscape are visually distinctive.
☑ The setup draws attention through movement and transformation.

☑ The projection mapping may raise curiosity about the experience, adn the tactile quality of the sand - its aligned with the gameplay as the core idea.

Initial engagement goals

Low Barrier
☑ The main interaction is intuitive: shape the sand and observe.
☑ The setup should minimise long explanations and technical friction.

☑ The facilitator is always close by as the experience is not room-scale, clearly showing you can contact them at any time.
☑ Affordances in game should feel intuitive. Affordances outside in the physical world may be more playful (sandbox + simple childlike play)

Deep engagement goals

Comparison
☑ Participants may compare different sand shapes and see different system states.
☑ Facilitators can encourage trying contrasting terrain forms.
☑There is no time-limit to the phases, meaning they can explore at their own pace.

Try
☑ The demo communicates what it feels like to use the technology through direct embodied interaction.

Immersion
☑ GodBox supports focused engagement by linking physical action and virtual transformation.
☑ The tactile quality of sand strengthens embodied immersion.
☑ Ease of entry will be difficult, and facilitators must let participants explore over telling them what to do next.

Resource efficiency, development

Platform
☑ GodBox appears modular, combining reusable hardware and interaction components.
☑ The demo can serve both research and public communication.
☐ The demo is not easily swapped of upgraded, as some of the components are delicate and implemented with specific SDKs, wrappers and software.

Fake
☑ If any aspect is simulated, that should reduce development effort without misleading the audience. (Switching phases)

Resource efficiency, operation

Independent Visitor
☑ Some parts of GodBox may be understandable independently, but the full XR experience still benefits from facilitation.
☐ It is not a self explanatory experience due to the phases in game. However, the participant has a large degree of autonomy once in.

Robustness
☐ The system may not survive repeated use by non-experts considering the tech setup (projector, realsense camera, sandbox, static computer, VR headset on link)
☑ Recovery from small failures should be quick .

GodBox weaknesses:
GodBox has significant issues with the independent visitor aspect and robustness, given the sequence of events in the experience requires “Wizard-of-Oz” phase changes and a short explanation of how to play. There are also many technological components involved, including the RealSense camera, projection mapping, sandbox sensing, VR headset connection, and Unity application. Because these systems depend on each other, failures in one component can disrupt the whole experience.

This trade-off was chosen because the goal of the project is to explore embodied interaction between a physical material (sand) and a virtual system, rather than to build a polished consumer product. The complexity allows the system to demonstrate an unusual interaction concept that would not be possible with a simpler setup.

Facilitators therefore play an important role in maintaining the experience. They help guide participants through the phases of the demo, handle technical transitions between system states, and repair alignment when breakdowns occur. While this reduces the independence of the system, it enables a more stable and meaningful demonstration of the research idea behind GodBox.

b) Written Discussion

b) Discussion

The alignment framework, preventative guidelines, and design principles influenced both how the GodBox demonstration is facilitated and how the experience itself was designed. Because GodBox combines a physical sandbox with XR interaction, the risk of misunderstanding, confusion, or technical disruption is relatively high. The frameworks from Petersson et al. and Bobbe et al. therefore help structure the experience in a way that reduces breakdowns while still allowing participants to explore the system in an open and playful way.

The preventative guidelines primarily influence how the demonstration begins. Informing sequentiality ensures that the participant understands the order of the experience before putting on the headset. This is particularly important in GodBox because the interaction unfolds in phases. Participants first explore how shaping the sand affects the virtual landscape, and later interact with menus to place objects and build a society. Explaining the sequence ahead of time helps participants orient themselves and prevents them from feeling lost once the headset is donned.

Informing about differences in interaction and technology is also important in this setup. Many participants approach XR demonstrations with expectations based on commercial VR systems that use controllers or familiar game interfaces. GodBox instead relies heavily on embodied interaction through the sandbox, and uses XR as the digital landscape that changes from that. Explaining how the realsense camera sense, and that the physical manipulation of sand drives the digital response helps align participant expectations with how the system actually works.

Preempting the physical surroundings is particularly relevant because GodBox exists (somewhat) simultaneously in a physical and virtual environment. Participants must understand where they should stand, how far they can move, and where hardware components such as the RealSense camera and projector are located. Establishing these boundaries beforehand reduces the likelihood of collisions or confusion once the participant's vision is occluded by the headset.

Negotiating touch also plays an important role in the facilitation of the demo. Because the participant cannot fully see their surroundings when wearing the headset, facilitators may need to guide them physically if they drift away from the sandbox or become disoriented, even though the sandbox should act as an anchoring point. Asking for consent beforehand establishes trust and ensures that participants understand why touch might occur during the demonstration.

During the demonstration itself, the five forms of alignment from Petersson et al. support the repair of breakdowns. Procedural alignment helps facilitators restate the current phase of the interaction and clarify what the participant should be doing. Technological alignment helps explain the capabilities and limitations of the system, particularly when gestures or interactions do not produce the expected result. Conversational alignment allows facilitators to maintain communication with the participant and check their understanding of what is happening. Visual alignment is supported by asking the participant to describe what they see and by monitoring the headset output on an external display/projected topography. Finally, physical alignment helps facilitators reposition the participant if necessary while maintaining comfort and safety.

Bobbe et al.'s design principles also shaped how the demonstration is structured. The sandbox provides a strong contextual metaphor that communicates the purpose of the system in an accessible

way. Shaping sand to create landscapes is a familiar activity that makes the experience easier to approach for people without technical expertise. The system also attempts to show invisible processes by making sensing and computation visible through changes in the virtual world and the projection on the sand.

Attraction and engagement principles are addressed through the tactile and visual qualities of the setup. The physical sandbox and the changing digital landscape projection act as an eye-catcher that attracts attention from spectators. Once participants begin interacting, the system encourages deeper engagement by allowing them to experiment with different terrain forms and observe how the world responds.

Finally, the framework highlights some limitations of the system. GodBox currently requires facilitators to guide participants through the phases and to manage technical transitions. This reduces the system's robustness and independence as a demonstration platform. However, these limitations are partly the result of the experimental nature of the prototype. The goal of the demonstration is not to present a finished product but to communicate the research concept of linking physical material interaction with virtual world-building.

Overall, the alignment framework, preventative guidelines, and design principles help structure the GodBox demonstration so that participants can understand the interaction, recover from confusion, and meaningfully engage with the system. They also help facilitators anticipate potential breakdowns and design the experience in a way that balances exploration with clarity.

c) Demo protocol for GodBox

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GodBox Demonstration Protocol

1. Welcome and framing

The facilitator welcomes the participant and introduces GodBox as an XR sandbox experience in which shaping the physical sand landscape affects a virtual world. The participant is told that the experience is exploratory and that there is no single correct way to interact with the system.

The participant is also asked whether they would like to use nitrile gloves during the experience in case of sensory or hygiene concerns.

2. Pre-demo briefing

The facilitator explains the sequence of the demonstration: introduction, headset donning, sandbox exploration, observation of changes in the virtual world, and the ending of the experience. The participant is informed how to start and stop the experience and what the facilitators will do during the demonstration.

3. Prior experience and expectations

The facilitator briefly asks whether the participant has used VR before. They explain that GodBox differs from typical VR systems because it is not primarily controller-based. Instead, the participant interacts with the experience by shaping the physical sand, which is sensed by a RealSense camera and translated into changes in the virtual landscape. The facilitator also demonstrates the basic gestures used to interact with the sand and resets the sand surface before the headset is donned.

4. Safety and consent

The facilitator shows the participant the sandbox, the correct standing position, and the safe movement area. They explain that light physical guidance may be used if necessary for safety or orientation, and ask the participant for consent before the experience begins. Voice-only guidance is offered as an alternative.

5. Interaction explanation

Before the headset is worn, the facilitator explains how the interaction works. If possible, a brief demonstration is given by shaping the sand and showing how the landscape changes on the projection or external display. The participant is told what kinds of responses to expect from the system and that some interactions may not work perfectly because the system is a prototype.

6. Headset donning and initial orientation

The facilitator assists the participant in putting on the headset and adjusting it if necessary. Once the headset is donned, the facilitator repeats the first step: orient to the scene, look toward the sandbox, and begin interacting slowly with the sand. The facilitator may ask the participant to briefly describe what they see in order to confirm visual alignment.

7. Main interaction phase

The participant explores the environment and shapes the sand landscape. The facilitator monitors the participant and the system output while maintaining safety and alignment. Minimal verbal prompts are used, such as open check-ins or confirmation questions, allowing the participant to explore the interaction at their own pace.

8. Breakdown repair if needed

If the participant becomes confused, silent, mispositioned, or disoriented, the facilitator repairs alignment using several strategies. This may include restating the current step (procedural alignment), clarifying how the system responds (technological alignment), asking the participant what they see (conversational and visual alignment), or gently guiding the participant's position if consent has been established (physical alignment).

9. Transition to ending

The facilitator marks the end of the interaction phase and informs the participant that the demonstration is concluding. The participant is guided to stop interacting and remove the headset with assistance if needed.

10. Debrief and reflection

After removing the headset, the participant is asked about their experience. Questions may focus on what they noticed about the connection between the sand and the virtual world, what felt intuitive or confusing, and what aspects of the experience were most engaging. This helps gather feedback and evaluate how effectively alignment was maintained during the demonstration.

Breakdown Prevention and Repair Plan

For GodBox, breakdowns can be prevented by clearly explaining the sequence of the demo before it begins, describing how the interaction differs from conventional VR, establishing the physical boundaries around the sandbox, and negotiating consent for potential physical guidance.

If a breakdown occurs, repair should begin with the least disruptive form of alignment. Facilitators first attempt conversational and procedural repair by restating instructions or asking clarifying questions. If confusion persists, technological explanations or visual reorientation may be used. Physical guidance is used only if necessary and with prior consent. This layered approach helps maintain immersion while ensuring participant safety and progress through the demonstration.